Liche Tsegaye (Li)

Open to Opportunities

Software Engineer & AI/ML Developer

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PROFESSIONAL SUMMARY

Software Engineer and 5th-year Software Engineering graduating student at Jimma University specializing in Artificial Intelligence, Retrieval-Augmented Generation (Agentic RAG), Python backend architecture, and full-stack development (MERN stack). Experienced in engineering autonomous multi-agent workflows, vector retrieval pipelines, and interactive voice systems. Proven communicator with high personal accountability and a dedication to writing clean, maintainable, production-ready software.

TECHNICAL SKILLS (DUAL SPECIALIZATION)

AI & Agentic RAG	Agentic RAG, LangChain, LlamaIndex, ChromaDB, FAISS, PyTorch, Google Gemini API, OpenAI APIs
Backend & Python	Python, FastAPI, Flask, Pydantic, Uvicorn, RESTful APIs, WebSockets, BeautifulSoup, Selenium
Full-Stack & Web	MERN Stack (MongoDB, Express.js, React, Node.js), JavaScript (ES6+), TypeScript, HTML5/CSS3
Data & DevOps	PostgreSQL, MongoDB, SQLite, Redis, Git, GitHub Actions, Docker, Linux, Render
Core Strengths	High Accountability, Extreme Ownership, Clear Technical Communication, Rapid Prototyping

WORK EXPERIENCE

Information Network Security Administration (INSA) — AI Engineering Intern

6 Months

RAG | Agentic RAG | LLM Orchestration | Vector DBs

- Engineered and scaled Retrieval-Augmented Generation (RAG) and Agentic RAG pipelines within the AI Department.
- Designed autonomous multi-agent routing systems incorporating intent detection, document evaluation, and contextual memory.
- Implemented hybrid retrieval combining dense vector embeddings with sparse keyword search and phonetic fuzzy matching.
- Collaborated on security-first AI pipelines, API rate-limit resilience, and round-robin multi-key failover managers.

FEATURED PROJECTS

Visit Ethiopia Agentic RAG System — Multi-Agent Tourism Intelligence

View Repo

Python | Agentic RAG | ChromaDB | BM25 Fuzzy | LangChain | Streamlit & Flask

- Architected an autonomous multi-agent system with a Fused Supervisor Planner handling intent classification, pronoun resolution, and search query expansion.
- Engineered specialized domain expert sub-agents for Itinerary Planning, Culture & Heritage (UNESCO sites), and Logistics & Visa rules.
- Integrated real-time currency conversion APIs and two-tier semantic query caching (≥ 0.82 similarity) returning zero-API-call responses on hits.

Li's AI Virtual Voice Assistant — Interactive Voice Portfolio

View Repo

FastAPI | WebSockets | Google Gemini AI | Guardrails | Local RAG Engine

- Engineered a full-duplex interactive voice portfolio assistant with WebSockets, client-side STT/TTS, and multi-key API failover.
- Built custom security guardrails preventing prompt injection attacks and sanitizing outgoing code leaks, system instructions, and filepaths.
- Developed an offline keyword RAG fallback ensuring uninterrupted visitor responses during quota exhaustion.

Password Exposure Checker — Zero-Knowledge Security Tool

View Repo

Python | SHA-1 | k-Anonymity | Have I Been Pwned API

- Engineered a credential security auditor querying Have I Been Pwned API using SHA-1 hashing and the mathematical k-Anonymity privacy model.
- Guaranteed zero credential leakage across the network by transmitting only the first 5 characters of the hash prefix.

Hacker News Signal Filter & Scraper — Content Aggregation & Curation

View Repo

Python | BeautifulSoup | Web Scraping | Data Filtering

- Dynamic multi-page web crawler aggregating top engineering discussions and filtering articles with scores > 99.
- Automated visual HTML report generation with vote-weighted ranking and respectful request throttling.

Document Automation & PDF Tool Suite — Programmatic Document Utilities

View Repo

Python | PyPDF2 | Automation | CLI

- Command-line automation scripts for batch PDF document merging, dynamic multi-page watermarking, and format validation.

Interactive JavaScript Games & Audio Suite — Browser Games & Audio Synthesis

View Games

JavaScript (ES6+) | DOM Manipulation | Web Audio API | Game Loops

- Developed a collection of responsive browser-based games including **Dice Game**, **Guess My Number**, and **World Quiz** featuring state machines and win-condition logic.
- Built an interactive virtual **Drum Kit** utilizing keyboard event listeners and asynchronous audio synthesis for zero-latency auditory feedback.

Batch Image Format Converter — Image Processing Utility

View Repo

Python | Pillow / PIL | Image Processing

- High-speed batch image converter transforming and optimizing JPG to lossless PNG formats with directory traversal.

EDUCATION & ACHIEVEMENTS

Jimma University — B.Sc. in Software Engineering

2015 E.C. – Present (5th Year Graduating Class)

- Active participant in university tech clubs, developer societies, and regional hackathons.
- Completed notable online technical certifications in modern AI/ML, backend engineering, and software architectures.